

Gold Price Prediction Using Commodity and Macroeconomic Indicators

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1. Background and Problem Statement

Gold is often viewed as a safe-haven asset and inflation hedge, but its price can be affected by many factors such as inflation, interest rates, the U.S. dollar, and other commodities. The problem we intend to explore is whether historical commodity and macroeconomic data can be used to predict future gold prices.

2. Objectives

- Analyze relationships between gold prices, inflation, interest rates, the U.S. dollar, and commodities such as silver, oil, copper, wheat, corn, and natural gas.
- Train and compare multiple regression models.
- Evaluate model performance.
- Discuss whether gold price prediction is realistic using this dataset, or whether the dataset is better suited for inflation/commodity relationship analysis.

3. Dataset

[Gold, Silver, Commodities vs Inflation](#)

The dataset contains historical time-series data from around 2000 to 2026, including prices of gold, silver, platinum, oil, copper, natural gas, wheat, and corn, along with CPI, inflation rate, interest rates, Treasury yields, the U.S. dollar index, and recession indicators.

Since this is an early topic idea, we may adjust the final focus after exploratory analysis. If direct gold price prediction is not effective.

Detecting Deepfake Voices Using Machine Learning

1. Background and Problem Statement

Advances in generative AI now allow extremely realistic synthetic voices that can mimic human speakers, raising risks for fraud, impersonation, and misinformation.

The problem I intend to solve is **Given a short audio clip of speech, automatically do binary classification whether the voice is real (human-recorded) or fake (AI-generated deepfake).**

2. Objectives

- Extract audio features (MFCCs, log-mel spectrograms) that capture spectral and temporal characteristics of speech.^{[7][11]}
- Train and evaluate at least one baseline classifier (e.g., SVM or logistic regression) and one simple deep model (e.g., CNN on spectrograms) for deepfake voice detection.
- Report metrics such as accuracy, precision, recall, F1-score, and confusion matrix, and discuss limitations and potential dataset artefacts.

4. Dataset

DEEP-VOICE: DeepFake Voice Recognition: <https://www.kaggle.com/datasets/birdy654/deep-voice-deepfake-voice-recognition>

It contains labeled audio samples of real human speech and AI-generated deepfake speech; files are organized into REAL and FAKE directories to support binary classification tasks.^{[8][5]}